
1. INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Anganwadi centers are essential for delivering nutrition and healthcare services to children, pregnant women, and lactating mothers. The primary objective is to ensure the timely and equitable distribution of nutritional food, enhancing the health outcomes and overall well-being of the community. However, the existing manual processes and paper-based records present significant challenges, highlighting the need for a more streamlined and automated approach to improve efficiency, accuracy, and transparency. Nutrition services are often disconnected from healthcare, resulting in gaps in care and inadequate support. Without automated systems, challenges remain in allocating resources, affecting the timely access to nutrition for beneficiaries. To address these challenges, our project 'NutriCare' aims to develop a digital platform that enables online candidate registration, efficient food distribution, and real-time monitoring of distribution. By utilizing technology, we seek to improve the health and well-being of children, pregnant women, and lactating mothers, while also enhancing the overall effectiveness of Anganwadi services. Our motivation is to create a positive impact on the lives of millions, ensuring every child, pregnant woman, and lactating mother receives nutritious food on time, empowering Anganwadi workers with technology to enhance their impact, and supporting the government's vision for a healthier India. We aim to bridge the nutrition gap for vulnerable populations and create a flexible model for digital transformation in healthcare.

1.2 THE PROPOSED SYSTEM

The proposed system for Anganwadi Nutrition Food Management aims to streamline and digitize the existing manual processes, ensuring efficient and transparent operations. This system introduces a digital platform that facilitates online candidate registration, efficient food distribution, and real-time monitoring of distribution. Admins can add and manage wards, anganwadi centers, supervisors, and teachers, as well as oversee food items and their quantities. Supervisors are empowered to allocate food items based on the number of registered candidates in each category, while teachers handle candidate registrations and update their physical details regularly. The system also includes authentication and password management to ensure that only authorized persons have access to perform their respective roles. Admins can generate detailed reports on various aspects of the system, helping them make better decisions

based on data. By utilizing technology, the proposed system aims to improve the health and well-being of children, pregnant women, and lactating mothers, thereby enhancing the overall effectiveness and transparency of Anganwadi services.

1.3 PROJECT SCOPE

1.3.1 LIMITATIONS OF EXISTING SYSTEM

- **Time Consuming:** The current system relies heavily on manual record-keeping, which is time-consuming and prone to delays. Staff must spend significant time filling out forms and managing paperwork.
- **Error-Prone:** Manual data entry increases the likelihood of errors in recording candidate details, food allocation, and distribution records.
- **Inconsistent Data:** Inconsistent methods of recording data can result in inaccuracies, making it difficult to maintain reliable records. Handwritten records are particularly susceptible to errors and omissions.
- **Delayed Updates:** In the existing system, updates to candidate records and inventory levels are often delayed, preventing real-time monitoring of the status of food distribution and candidate health.
- **Limited Reporting:** Generating comprehensive reports on food allocation, distribution is challenging and time-consuming, limiting the ability to make informed decisions.
- **Inaccurate Allocation:** Without a systematic approach, food allocation is often based on estimates rather than accurate data, leading to over-allocation or under-allocation of resources.
- **Lack of Accountability:** The absence of a centralized system makes it difficult to track the actions of different stakeholders (admins, supervisors, teachers), leading to a lack of accountability and potential misuse of resources.
- **Transparency Issues:** Manual processes and paper-based records do not provide transparency in the allocation and distribution process, making it hard to ensure fair and equitable distribution of resources.
- **Data Security:** Physical records are not secure and can be easily accessed by unauthorized individuals, resulting in leaking of sensitive information.
- **Privacy Issues:** Personal information of candidates is vulnerable to unauthorized access and misuse, compromising the privacy and confidentiality of their data.

- **Adaptability to Modern Needs:** Inability to adapt quickly to changing nutritional guidelines, technological advancements, and evolving healthcare requirements due to rigid manual systems.

1.3.2 ADVANTAGES OF PROPOSED SYSTEM

- **Increased Efficiency:** Automation of candidate registration, food allocation reduces manual errors and speeds up operations.
- **Enhanced Transparency:** Real-time data tracking and reporting improve transparency in resource management and distribution, ensuring that food supplies are allocated equitably among all beneficiaries.
- **Improved Data Accuracy:** Digital data entry and management minimize errors, ensuring that information about candidates and food supplies is accurate and reliable. This enhances the quality of decision-making and program evaluation.
- **Real-Time Monitoring:** The system allows for real-time monitoring of food stocks, distribution rates, and beneficiary participation.
- **Scalability:** The digital platform can easily scale to accommodate increasing numbers of beneficiaries or changing nutritional needs across different regions.
- **Resource Optimization:** Accurate assessment and allocation of resources based on real-time data reduce wastage and shortages.
- **Enhanced Accountability:** Digital records provide a clear audit trail, improving accountability and reducing the risk of misuse or inefficiencies in food supply management.
- **Better Health Outcomes:** Timely and accurate provision of nutritious food supports the health and well-being of children, pregnant women, and lactating mothers.
- **Cost-Effective:** While there are initial setup costs, the long-term savings from reduced manual labor, errors, and inefficiencies make the system cost-effective.
- **Adaptability:** The system can quickly adapt to changes in nutritional guidelines, policies, and technological advancements, ensuring ongoing relevance and effectiveness.
- **Enhanced Reporting:** Comprehensive and real-time reports provide valuable insights for program evaluation, planning, and decision-making. This supports continuous improvement of Anganwadi services.

2. SYSTEM ANALYSIS

2.1 INTRODUCTION

Software Engineering is the analysis, design, construction, verification and management of technical or social entities. To engineer software accurately, a software engineering process must be defined. System analysis is a detailed study of the various operations performed by the system and their relationship within and module of the system. It is a structured method for solving the problems related to the development of a new system. The detailed investigation of the present system is the focal point of system analysis. This phase involves the study of parent system and identification of system objectives. Information has to be collected from all people who are affected by or who use the system. During analysis, data are collected on the variable files, decision point and transactions handled by the present system. The main aim of system is to provide the efficient and user friendly automation. So the system analysis process should be performed with extreme precision, so that an accurate picture of existing system, its disadvantages and the requirements of the new system can be obtained.

System analysis involves gathering the necessary information and using the structured tool for analysis. This includes the studying existing system and its drawback, designing a new system and conducting cost benefit analysis. System analysis is a problem solving activity that requires intensive communication between the system users and system developers. The system is studied to the minute detail and analyzed. The system is viewed as a whole and the inputs to the system are identified. The outputs from the organization are traced through various phases of processing of inputs.

There are a number of different approaches to system analysis. When a computer based information system is developed, systems analysis (according to the Waterfall model) would constitute the following steps:

- The development of a feasibility study, involving determining whether a project is economically, technologically and operationally feasible.
- Conducting fact-finding measures, designed to ascertain the requirements of the system's end-users. These typically span interviews, questionnaires, or visual observations of work on the existing system.
- Gauging how the end-users would operate the system (in terms of general experience in using computer hardware or software), what the system would be used for and so on.

Techniques such as interviews, questionnaires etc. can be used for the detailed study of these processes. The data collected by these sources must be scrutinized to arrive at a conclusion.

The conclusion is an understanding of how the system functions. This system is called the Existing System. The Existing system is then subjected to close observation and the problem areas are identified. The designer now functions as a problem solver and tries to sort out the difficulties that the enterprise faces. The solutions are given as a proposal which is the Proposed System. The proposal is then weighed with the existing system analytically and the best one is selected. The proposal is then presented to the user for an endorsement by the user. The proposal is reviewed on user request and suitable changes are made. This is a loop that ends as soon as the user is satisfied with the proposal.

2.2 STAKEHOLDERS IN THIS PROJECT

2.2.1 ADMIN

The admin, represented by the Panchayat, holds a pivotal role in the Anganwadi Nutrition Food Supply System, managing and overseeing various critical aspects to ensure efficient service. The admin is responsible for the system's infrastructure, including site content, templates, style sheets, and processes, ensuring the design and functionality meet the required standards. They manage registration for supervisors, teachers, various group wise categories, food items and Anganwadi, assigning appropriate access rights and permissions. Content management is a key responsibility, involving the addition, updating, and deletion of site content to maintain accuracy and relevance. The admin oversees the equitable allocation of food resources to Anganwadi centers based on real-time data, monitors the activities of supervisors and teachers. The admin generates reports and analyzes data for system performance evaluation and makes informed decisions for continuous improvement. Overall, the admin ensures that the nutritional needs of children, pregnant women, and lactating mothers are met effectively, contributing to the community's health and well-being.

2.2.2 SUPERVISOR

The supervisor plays a critical role, acting as a link between the administrative body (Panchayat) and the Anganwadi centers. The supervisor is responsible for managing and overseeing the daily operations within their assigned ward, ensuring the efficient allocation and distribution of nutritional supplies to the Anganwadi centers under their

management. They manage the allocation of food items to each Anganwadi center, ensuring that the supplies are distributed based on the specific needs and categories of candidates (children, pregnant women, lactating mothers). This involves calculating the required quantities of food based on the number of registered candidates in each category and coordinating the timely delivery of these supplies to the respective centers. They monitor the activities within the Anganwadi centers, ensuring that the food supplies are utilized appropriately and that the nutritional needs of the candidates are met effectively. Inshort, Supervisor ensures that nutritional supplies are effectively managed and distributed.

2.2.3 TEACHER

In the Anganwadi, teacher register children, pregnant women, and lactating mothers. Each month, they update and store the height and weight details of children, pregnant women, and lactating mothers. This data is essential for accurately categorizing candidates and determining their nutritional needs. Teachers ensure that the right food is allocated to each candidate based on these updates, ensuring transparency and accountability. The teacher also updates the stock regularly to maintain an accurate inventory, ensuring timely reordering and preventing shortages.

2.3 SOFTWARE REQUIREMENT SPECIFICATION

2.3.1 ADMIN

1. The system should have a login facility using a username and password.
2. The system should allow the admin to view the homepage.
3. The system should provide the facility to add categories such as children, pregnant women, and lactating women, and allow adding further categories if needed.
4. There should be provision to add, view, edit ward.
5. There should be provision to add, view, edit Anganwadi under each ward.
6. Admin should have the facility to add, view, edit Supervisor for each ward.
7. Admin should have the facility to add, view, edit Teacher for each Anganwadi.
8. Admin should have the permission to add food item and set fixed quantities for each category.
9. Admin should have the facility to view food items and modify food items.
10. The system should provide the facility for the admin to log out.

2.3.2 SUPERVISOR

1. System should allow the supervisor to login to the site using username and password.
2. The system should provide the facility for the supervisor to change their password.
3. System should have the facility to view the homepage of Supervisor.
4. The system should provide the facility to allocate the food items to each Anganwadi according to number of candidates registered in each category .
5. System should have the facility to view the updated stock details from teacher's side.
6. The system should provide the facility for the supervisor to logout.

2.3.3 TEACHER

1. The system should provide the facility for the teacher to log in using a username and password.
2. System should provide facility to change password.
3. System should have the facility to view the homepage of teacher.
4. Teacher should have the facility to view the categories.
5. Teacher should have the provision to register new candidates into the corresponding category under their respective Anganwadi centers.
6. The system should provide the facility to view candidates in each category.
7. The system should provide the facility to update the physical details of candidates while distributing food and update the system accordingly.
8. The system should provide the facility to view stock details.
9. The system should allow the teacher to update the stock when distributing food to individuals.
10. The system should provide the facility for the teacher to log out.

Table 2.1 Sign off table

Sl.No.	Name & Designation	Date	Accepted (Yes/No)
1	Deepa M Assistant Professor Santhigiri College of Computer Sciences		
2	Janaky Babu Developer		
3	Keerthana S Developer		

2.4 FEASIBILITY STUDY

Feasibility is defined as the practical extent to which a project can be performed successfully. To evaluate feasibility, a feasibility study is performed, which determines whether the solution considered to accomplish the requirements is practical and workable in the software. Information such as resource availability, cost estimation for software development, benefits of the software to the organization after it is developed and cost to be incurred on its maintenance are considered during the feasibility study. The objective of the feasibility study is to establish the reasons for developing the software that is acceptable to users, adaptable to change and conformable to established standards. Various other objectives of feasibility study are listed below.

- To analyse whether the software will meet organizational requirements.
- To determine whether the software can be implemented using the current technology and within the specified budget and schedule.
- To determine whether the software can be integrated with other existing software. When our project guide as well as our client Ms. Deepa M told us regarding the mini project and about Word to the Wise for getting the desired product developed, it comes up with rough idea about what all functions the software must perform and which all features are expected from the software.

Referencing to this information, we does a studies and discussions about whether the desired system and its functionality are feasible to develop and the output of this phase is a feasibility study report that should contained adequate comments and recommendations.

Various types of feasibility that we checked include technical feasibility, operational feasibility, and economic feasibility.

TECHNICAL FEASIBILITY

Technical feasibility assesses the current resources (such as hardware and software) and technology, which are required to accomplish user requirements in the software within the allocated time and budget. For this, the software development team ascertains whether the current resources and technology can be upgraded or added in the software to accomplish specified user requirements. Technical feasibility also performs the following tasks.

- Analyses the technical skills and capabilities of the software development team members.
- Determines whether the relevant technology is stable and established.
- Ascertains that the technology chosen for software development has a large number of users so that they can be consulted when problems arise or improvements are required.

From our perspective there are two languages PHP, HTML and database MySQL which are used to develop this web based applications. PHP is used in the front end and MySQL is used in the back end. The Word to the Wise is web based and thus can be accessed through any browsers. As we are using these latest technologies which are currently trending and used by a number of developers across the globe, we can say that our project is technically feasible.

OPERATIONAL FEASIBILITY

Operational feasibility assesses the extent to which the required software performs a series of steps to solve business problems and user requirements. This feasibility is dependent on human resources (software development team) and involves visualizing whether the software will operate after it is developed and be operative once it is installed. Operational feasibility also performs the following tasks.

- Determines whether the problems anticipated in user requirements are of high priority.
- Determines whether the solution suggested by the software development team is acceptable.
- Analyses whether users will adapt to a new software.

- Determines whether the organization is satisfied by the alternative solutions proposed by the software development team.

We found that our project will be satisfied for the client since we were discussing every detail about the software with the client at every step. The most important part of operational feasibility study is the input from client. So the software is built completely according to the requirements of the client. We have used the current industry standards for the software. Hence we can say that this software is operationally feasible.

ECONOMIC FEASIBILITY

Economic feasibility determines whether the required software is capable of generating financial gains for an organization. It involves the cost incurred on the software development team, estimated cost of hardware and software, cost of performing feasibility study, and so on. For this, it is essential to consider expenses made on purchases (such as hardware purchase) and activities required to carry out software development. In addition, it is necessary to consider the benefits that can be achieved by developing the software. Software is said to be economically feasible if it focuses on the issues listed below.

- Cost incurred on software development to produce long-term gains for an organization.
- Cost required to conduct full software investigation (such as requirements elicitation and requirements analysis).
- Cost of hardware, software, development team, and training.

It is estimated that our project is economically feasible as development cost is very minimal since the tools and technologies used are available online. It's a group student project so there are no personnel costs. Development time is well planned and will not affect other operations and activities of the individuals. Once the system has been developed, the companies purchasing the system will be providing with a manual for training purposes. There is no need to purchase new hardware since the existing computers can still be used to implement the new system.

2.5 SOFTWARE DEVELOPMENT LIFECYCLE MODEL

One of the basic notions of the software development process is SDLC models which stand for Software Development Life Cycle models. SDLC – is a continuous process, which starts from the moment, when it's made a decision to launch the project, and it ends at the moment of its full remove from the exploitation. Software development

lifecycle (SDLC) is a framework that defines the steps involved in the development of software. It covers the detailed plan for building, deploying and maintaining the software. SDLC defines the complete cycle of development i.e. all the tasks involved in gathering a requirement for the maintenance of a Product.

Some of the common SDLC models are Waterfall Model, V-Shaped Model, Prototype Model, Spiral Model, Iterative Incremental Model, Big Bang Model, Agile Model. We used Agile Model for our Project.

AGILE MODEL

Agile Model is a combination of the Iterative and incremental model. This model focuses more on flexibility while developing a product rather than on the requirement. In the agile methodology after every development iteration, the client is able to see the result and understand if he is satisfied with it or he is not. Extreme programming is one of the practical use of the agile model. The basis of this model consists of short meetings where we can review our project. In Agile, a product is broken into small incremental builds. It is not developed as a complete product in one go. At the end of each sprint, the project guide verifies the product and after his approval, it is finalised. Client feedback is taken for improvement and his suggestions and enhancement are worked on in the next sprint. Testing is done in each sprint to minimize the risk of any failures.

Advantages of Agile Model:

- It allows more flexibility to adapt to the changes.
- The new feature can be added easily.
- Customer satisfaction as the feedback and suggestions are taken at every stage.
- Risks are minimized thanks to the flexible change process.

Disadvantages :

- Lack of documentation.
- If a customer is not clear about how exactly they want the product to be, then the project would fail.
- With all the corrections and changes there is possibility that the project will exceed expected time.

2.6 HARDWARE AND SOFTWARE REQUIREMENTS

2.6.1 SOFTWARE SPECIFICATION

This project is built upon the latest technology software.

Front end	:	HTML, JavaScript
Development tool	:	PHP
Database	:	MySQL
Web server	:	WAMP server
Operating System	:	Windows 11

2.6.1.1 PHP

PHP is a server-side scripting language designed for web development but also used as a general-purpose programming language. As of January 2013, PHP was installed on more than 240 million websites (39% of those sampled) and 2.1 million web servers. Originally created by Rasmus Lerdorf in 1994, the reference implementation of PHP is now produced by The PHP Group. While PHP originally stood for Personal Home Page, it now stands for PHP: Hypertext Pre-processor, a recursive acronym.

PHP code can be simply mixed with HTML code, or it can be used in combination with various templating engines and web frameworks. PHP code is usually processed by a PHP interpreter, which is usually implemented as a web server's native module or a Common Gateway Interface (CGI) executable. After the PHP code is interpreted and executed, the web server sends resulting output to its client, usually in form of a part of the generated web page - for example, PHP code can generate a web page's

HTML code, an image, or some other data. PHP has also evolved to include a command-line interface (CLI) capability and can be used in standalone graphical applications.

PHP is free software released under the PHP License. PHP has been widely ported and can be deployed on most web servers on almost every operating system and platform, free of charge

2.6.1.2 MySQL

MySQL is the world's most popular open source database software, with over 100 million copies of its software downloaded or distributed throughout its history. With its superior speed, reliability, and ease of use, MySQL has become the preferred choice

for Web, Web 2.0, SaaS, ISV, Telecom companies and forward-thinking corporate IT Managers because it eliminates the major problems associated with downtime, maintenance and administration for modern, online applications.

Many of the world's largest and fastest-growing organizations use MySQL to save time and money powering their high-volume Web sites, critical business systems, and packaged software — including industry leaders such as Yahoo!, Alcatel-Lucent, Google, Nokia, YouTube, Wikipedia, and Booking.com.

The flagship MySQL offering is MySQL Enterprise, a comprehensive set of production- tested software, proactive monitoring tools, and premium support services available in an affordable annual subscription.

MySQL is a key part of LAMP (Linux, Apache, MySQL, PHP / Perl / Python), the fast-growing open source enterprise software stack. More and more companies are using LAMP as an alternative to expensive proprietary software stacks because of its lower cost and freedom from platform lock-in.

MySQL was originally founded and developed in Sweden by two Swedes and a Finn: David Axmark, Allan Larsson and Michael "Monty" Widenius, who had worked together since the 1980's. MySQL, the most popular Open Source SQL database management system, is developed, distributed, and supported by Oracle Corporation.

MySQL is a database management system. A database is a structured collection of data. It may be anything from a simple shopping list to a picture gallery or the vast amounts of information in a corporate network. To add, access, and process data stored in a computer database, you need a database management system such as MySQL Server. Since computers are very good at handling large amounts of data, database management systems play a central role in computing, as standalone utilities, or as parts of other applications.

MySQL databases are relational. A relational database stores data in separate tables rather than putting all the data in one big storeroom. The database structures are organized into physical files optimized for speed. The logical model, with objects such as databases, tables, views, rows, and columns, offers a flexible programming environment. You set up rules governing the relationships between different data fields, such as one-to one, one-to- many, unique, required or optional, and —pointers between different tables. The database enforces these rules, so that with a well-designed

database, your application never sees inconsistent, duplicate, orphan, out-of-date, or missing data.

The SQL part of —MySQL stands for —Structured Query Language. SQL is the most common standardized language used to access databases. Depending on your programming environment, you might enter SQL directly (for example, to generate reports), embed SQL statements into code written in another language, or use a language-specific API that hides the SQL syntax.

SQL is defined by the ANSI/ISO SQL Standard. The SQL standard has been evolving since 1986 and several versions exist. In this manual, —SQL-92 refers to the standard released in 1992, —SQL:1999 refers to the standard released in 1999, and —SQL:2003 refers to the current version of the standard.

We use the phrase —the SQL standard to mean the current version of the SQL Standard at any time. MySQL software is Open Source. Open Source means that it is possible for anyone to use and modify the software. Anybody can download the MySQL software from the Internet and use it without paying anything. If you wish, you may study the source code and change it to suit your needs.

The MySQL software use the GPL (GNU General Public License), <http://www.fsf.org/licenses/>, to define what you may and may not do with the software in different situations. If you feel uncomfortable with the GPL need to embed MySQL code into a commercial application, you can buy a commercially licensed version from us.

The MySQL Database Server is very fast, reliable, scalable, and easy to use. If that is what you are looking for, you should give it a try. MySQL Server can run comfortably on a desktop or laptop, alongside your other applications, web servers, and so on, requiring little or no attention. If you dedicate an entire machine to MySQL, you can adjust the settings to take advantage of all the memory, CPU power, and I/O capacity available. MySQL can also scale up to clusters of machines, networked together.

MySQL Server was originally developed to handle large databases much faster than existing solutions and has been successfully used in highly demanding production environments for several years. Although under constant development, MySQL Server today offers a rich and useful set of functions. Its connectivity, speed, and security make MySQL Server highly suited for accessing databases on the Internet. MySQL Server works in client/server or embedded systems.

The MySQL Database Software is a client/server system that consists of a multi-threaded SQL server that supports different backends, several different client programs and libraries, administrative tools, and a wide range of application programming interfaces (APIs). We also provide MySQL Server as an embedded multi-threaded library that you can link into your application to get a smaller, faster, easier-to-manage standalone product.

A large amount of contributed MySQL software is available. MySQL Server has a practical set of features developed in close cooperation with our users. It is very likely that your favourite application or language supports the MySQL Database Server.

2.6.1.3 WAMP SERVER

WAMP Server is a Windows web development environment. It allows you to create web applications with Apache2, PHP and a MySQL database. Alongside, PhpMyAdmin allows you to manage easily your databases. WAMP Server refers to a software stack for the Microsoft Windows operating system, created by Romain Bourdon and consisting of the Apache web server, Open SSL for SSL support, MySQL database and PHP programming language. WAMP Server is a Web development platform on Windows that allows you to create dynamic Web applications with Apache2, PHP, MySQL and MariaDB. WampServer automatically installs everything you need to intuitively develop Web applications. You will be able to tune your server without even touching its setting files. Best of all, WampServer is available for free (under GPML license) in both 32 and 64 bit versions. Wampserver is not compatible with Windows XP, SP3, or Windows Server 2003.

WAMP Server's functionalities are very complete and easy to use so we won't explain here how to use them.

With a left click on WAMP Server's icon, you will be able to:

- manage your Apache and MySQL services.
- switch online/offline (give access to everyone or only localhost)
- install and switch Apache, MySQL and PHP releases
- manage your server's settings
- access your logs
- access your settings files
- create alias

2.6.1.4 VISUAL STUDIO

Visual Studio Code is a source-code editor that can be used with a variety of programming languages, including C, C#, C++, FORTRAN, Go, Java, JavaScript, Node.js, Python, Rust, and Julia. It is based on the Electron framework, which is used to develop Node.js web applications that run on the Blink layout engine. Visual Studio Code employs the same editor component (code named "Monaco") used in Azure DevOps..

Out of the box, Visual Studio Code includes basic support for most common programming languages. This basic support includes syntax highlighting, bracket matching, code folding, and configurable snippets. Visual Studio Code also ships with Intelligent for JavaScript, Type Script, JSON, CSS, and HTML. Support for additional languages can be provided by freely available extensions on the VS Code Marketplace. Instead of a project system, it allows users to open one or more directories, which can then be saved in work spaces for future reuse. This allows it to operate as a language-agnostic code editor for any language. It supports many programming languages and a set of features that differs per language. Unwanted files and folders can be excluded from the project tree via the settings. Many Visual Studio Code features are not exposed through menus or the user interface but can be accessed via the command palette.

2.6.1.5 WINDOWS 11

Windows 11 represents a substantial evolution from its predecessor, Windows 10, with a focus on modernizing the user experience and enhancing productivity. Central to its redesign is a sleek and intuitive interface, featuring a centered Start menu and taskbar with rounded corners, providing a cohesive and visually appealing environment. The introduction of Snap Layouts and Snap Groups revolutionizes multitasking by enabling users to effortlessly organize and switch between applications across multiple windows and virtual desktops. The Microsoft Store has undergone a significant overhaul, now offering a curated selection of apps that includes both traditional Win32 applications and modern Universal Windows Platform (UWP) apps, with support for third-party commerce platforms to empower developers and expand app availability. Gamers will appreciate the integration of Auto HDR, DirectStorage, and DirectX 12 Ultimate, promising improved graphics fidelity, faster load times, and enhanced gaming experiences. For seamless communication, Windows 11 integrates Microsoft Teams directly into the taskbar, enabling quick access to chat, voice, and video calls. Security

enhancements, such as updated system requirements for enhanced hardware security and Windows Hello for biometric authentication, underscore Microsoft's commitment to protecting user data and privacy. Furthermore, advancements in touch, pen, and voice input ensure a responsive and versatile user experience, catering to a range of interaction preferences. Other new features include support for running Android apps through the Amazon Appstore, a revamped Microsoft Store with a broader range of applications, improved gaming performance through Auto HDR and DirectStorage, and enhanced security features such as Windows Hello for faster, more secure login options. Initial reviews of Windows 11 have been generally positive, praising its design changes, performance improvements, and new features. Other new features include support for running Android apps through the Amazon Appstore, a revamped Microsoft Store with a broader range of applications, improved gaming performance through Auto HDR and Direct Storage, and enhanced security features such as Windows Hello for faster, more secure login options. Initial reviews of Windows 11 have been generally positive, praising its design changes, performance improvements, and new features. Critics have noted concerns over hardware requirements and compatibility issues, particularly with older hardware, as well as some controversial decisions such as the requirement for a Microsoft account and internet connection during setup. Despite these criticisms, Windows 11 represents Microsoft's continued evolution of its operating system, aiming to enhance productivity, security, and user experience across a range of devices. Overall, Windows 11 emerges as a refined and feature-rich operating system that balances aesthetics, productivity, and security in a rapidly evolving digital landscape.

2.6.1.6 MICROSOFT WORD

Microsoft Word (or simply Word) is a word processor developed by Microsoft. It was first released on October 25, 1983 under the name Multi-Tool Word for Xenix systems. Subsequent versions were later written for several other platforms including IBM PCs running DOS (1983), Apple Macintosh running the Classic Mac OS (1985), AT&T Unix PC (1985), Atari ST (1988), OS/2 (1989), Microsoft Windows (1989), SCO Unix (1994), and macOS (formerly OS X; 2001).

Commercial versions of Word are licensed as a standalone product or as a component of Microsoft Office, Windows RT or the discontinued Microsoft Works suite. Unlike most MS-DOS programs at the time, Microsoft Word was designed to be used with a mouse. Advertisements depicted the Microsoft Mouse, and described Word as a

WYSIWYG, windowed word processor with the ability to undo and display bold, italic, and underlined text, although it could not render fonts. It was not initially popular, since its user interface was different from the leading word processor at the time, WordStar. However, Microsoft steadily improved the product, releasing versions 2.0 through 5.0 over the next six years. In 1985, Microsoft ported Word to the classic Mac OS (known as Macintosh System Software at the time). This was made easier by Word for DOS having been designed for use with high-resolution displays and laser printers, even though none were yet available to the general public. Following the precedents of LisaWrite and MacWrite, Word for Mac OS added true WYSIWYG features. It fulfilled a need for a word processor that was more capable than MacWrite. After its release, Word for Mac OS's sales were higher than its MS-DOS counterpart for at least four years.

2.6.1.7 SMARTDRAW

SmartDraw is a diagram tool used to make flowcharts, organization charts, mind maps, project charts, and other business visuals. SmartDraw has two versions: an online edition and a downloadable edition for Windows desktop. SmartDraw integrates with Microsoft Office products including Word, PowerPoint, and Excel and G Suite applications like Google Docs and Google Sheets. SmartDraw has apps for Atlassian's Confluence, Jira, and Trello. SmartDraw is compatible with Google Drive, Dropbox, Box, and OneDrive.

Since 1994, the mission of SmartDraw Software has been to expand the ways in which people communicate so that we can clearly understand each other, make informed decisions, and work together to improve our businesses and the world. We accomplish this by creating software and services that make it possible for people to capture and present information as visuals, while being a pleasure to use. In 2019, we took this to the next level by launching VisualScript, which makes it easy to visualize data in relational formats like trees, flows, and timelines, automatically, without any human input. VisualScript is a relationship visualization platform that empowers organizations to visualize data across siloed ecosystems and gain critical insights in real-time. Today, SmartDraw Software is one of the most sophisticated digital marketing organizations in the world with over 90,000 unique visitors to our website each business day and in excess of 3,000,000 installations of our apps each year. SmartDraw is used by more than half of the Fortune 500 and by over 250,000 public and private enterprises of all

sizes around the world. Privately held, SmartDraw Software is headquartered in San Diego, California.

2.6.2 HARDWARE REQUIREMENTS

The selection of hardware configuring is a very task related to the software development, particularly inefficient RAM may affect adversely on the speed and corresponding on the efficiency of the entire system. The processor should be powerful to handle all the operations. The hard disk should have the sufficient to solve the database and the application.

Hardware used for development:

CPU	: Intel i5 Processor
Memory	: 4 GB
Cache	: 6 MB
Hard Disk	: 1 TB
Monitor	: 15.6" Monitor
Keyboard	: Standard 108 keys Enhanced Keyboard
Mouse	: Optical Mouse

Minimum Hardware Required For Implementation:

CPU	: Pentium IV Processor
Memory	: 256MB Above
Cache	: 512 KB Above
Hard Disk	: 20 GB Above
Monitor	: Any
Keyboard	: Any
Mouse	: Any

3. SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

A system architecture or system's architecture is the conceptual model that defines the structure, behaviour, and more views of a system. An architecture description is a formal description and representation of a system, organized in a way that supports reasoning about the structures of the system. System architecture can comprise system components, the externally visible properties of those components, the relationships (e.g. the behaviour) between them. It can provide a plan from which products can be procured, and systems developed, that will work together to implement the overall system. There have been efforts to formalize languages to describe system architecture; collectively these are called architecture description languages (ADLs).

The system architecture can best be thought of as a set of representations of an existing (or to be created) system. It is used to convey the informational content of the elements comprising a system, the relationships among those elements, and the rules governing those relationships. The architectural components and set of relationships between these components that architecture describes may consist of hardware, software, documentation, facilities, manual procedures, or roles played by organizations or people. System architecture is primarily concerned with the internal interfaces among the system's components or subsystems, and the interface between the system and its external environment, especially the user.

The structural design reduces complexity, facilitates change and result in easier implementation by encouraging parallel development of different parts of the system. The procedural design transforms structural elements of program architecture into a procedural description of software components. The architectural design considers architecture as the most important functional requirement. The system is based on the three-tier architecture.

The first level is the user interface (presentation logic), which displays controls, receives and validates user input. The second level is the business layer (business logic) where the application specific logic takes place. The third level is the data layer where the application information is stored in files or database. It contains logic about to retrieve and update data. The important feature about the three-tier design is that information only travels from one level to an adjacent level.

3.2 MODULE DESIGN

Modular programming is a software design technique that emphasizes separating the functionality of a program into independent, interchangeable modules, such that each contains everything necessary to execute only one aspect of the desired functionality. Conceptually, modules represent a separation of concerns, and improve maintainability by enforcing logical boundaries between components.

Different modules of this project include.

1. AUTHENTICATION

This module ensures that only authorized persons can log in and perform actions based on their roles. Admin can log into the system by using their corresponding username and password. Admin can perform all the actions in system only after logging in. Supervisor and Teacher login into the system by using their respective username and password, allowing them to perform their specific tasks. Additionally, this module supports password management functionalities such as changing passwords and reset their passwords if forgotten. This ensures that user accounts remain secure and accessible only to authorized individuals.

2. MASTER ENTRIES

The Master Entries module allows administrators to manage the data of the system. This includes adding, viewing, and editing categories such as children, pregnant women and lactating women. Admin can also manage wards by adding new wards, viewing the existing ones, and also edit it. Furthermore, admins can add and manage Anganwadi centers under each ward. They can also add and manage supervisors and teachers, ensuring each center has the appropriate staff. Additionally, the admin can add, view, and edit food items, assigning fixed quantities to each category. Also, teachers can register new candidates into the corresponding categories under their respective Anganwadi centers.

3. ANGANWADI ACTIVITIES

This module focuses on managing the food supply and ensuring that stock levels are accurately recorded and updated. Supervisors can allocate food items to each Anganwadi center based on the number of registered candidates in each category. They can also view updated stock details reported by teachers to monitor food distribution effectively. Teachers can update the stock records when distributing food to candidates and update physical details such as height

and weight during food collection. This module ensures that stock levels are maintained accurately and that food distribution is tracked in real-time.

4. GENERATE REPORT

This module allows the admin to view the details of the site through various reports. This includes pie chart, graph and some table information. The reports provide the valuable information from the system. In our system all reports are important because the system eliminate the use of manipulation of the paper work. The system generates useful information's. Admins can generate reports on candidate registrations, including monthly updates on physical details like height and weight, food distribution reports detail allocations to each Anganwadi center etc so that the Admin can easily manage the website. This will helpful for the admin to generate useful information. The reports show the significance of the system.

3.3 DATABASE DESIGN

A database is a collection of interrelated data stored with minimum redundancy to serve many users quickly and efficiently. The general objective is to make information access easy, quick, inexpensive and flexible for the users. The general theme behind a database is to integrate all information. Database design is recognized as a standard of management information system and is available virtually for every computer system. In database design several specific objectives are considered:

- Ease of learning and use
- Controlled redundancy
- Data independence
- More information at low cost
- Accuracy and integrity
- Recovery from failure
- Privacy and security
- Performance

A database is an integrated collection of data and provides centralized access to the data. Usually, the centralized data managing the software is called RDBMS. The main significant difference between RDBMS and other DBMS is the separation of data as seen by the program and data has in direct access to stores device. This is the difference between logical and physical data.

3.3.1 NORMALIZATION

Designing a database is complete task and the normalization theory is a useful aid in the design process. The process of normalization is concerned with transformation of conceptual schema into computer representation form. There will be need for most databases to grow by adding new attributes and new relations. The data will be used in new ways. Tuples will be added and deleted. Information stored may undergo updating also. New association may also be added. In such situations the performance of a database is entirely depend upon its design. A bad database design may lead to certain undesirable things like:

- Repetition of information
- Inability to represent certain information
- Loss of information

To minimize these anomalies, Normalization may be used. If the database is in a normalized form, the data can be growing without, in most cases, forcing the rewriting application programs. This is important because of the excessive and growing cost of maintaining an organization's application programs and its data from the disrupting effects of database growth. As the quality of application programs increases, the cost of maintaining the without normalization will rise to prohibitive levels. A normalized database can also encompass many related activities of an organization thereby minimizing the need for rewriting the applications of programs. Thus, normalization helps one attain a good database design and there by ensures continued efficiency of database.

Normalization theory is built around the concept of normal forms. A relation is said to be in normal form if it satisfies a certain specified set of constraints. For example, a relation is said to be in first normal form (1NF) if it satisfies the constraint that it contains atomic values only. Thus, every normalized relation is in 1NF. Numerous normal forms have been defined. Codd defined the first three normal forms.

All normalized relations are in 1NF, some 1NF relations are also in 2NF and some 2NF relations are also in 3NF. 2NF relations are more desirable than 1NF and 3NF are more desirable than 2NF. That is, the database designer should prefer 3NF than 1NF or 2NF. Normalization procedure states that a relation that is in some given normal form can be converted into a set of relations in a more desirable form. We can define this procedure as the successive reduction of a given collection of relations to some more

desirable form. This procedure is reversible. That is, it is always possible to take the output from the procedure and convert them back into input. In this process, no information is lost. So it is also called “no loss decomposition”.

First Normal Form

A relation is in first normal form (1NF) if and all its attributes are based on single domain. The objective of normalizing a table is to remove its repeating groups and ensure that all entries of the resulting table have at most single value.

Second Normal Form

A table is said to be second Normal Form (2NF), when it is in 1NF and every attribute in record is functionally dependent upon the whole key, and not just a part of the key.

Third Normal Form

A table is in third Normal Form (3NF), when it is in 2NF and every non-key attribute is functionally dependent on just the primary key.

3.3.2 TABLE STRUCTURE

Table is a collection of complete details about a particular subject. These data are saved in rows and Columns. The data of each Row are different units. Hence, rows are called RECORDS and Columns of each row are called FIELDS.

Data is stored in tables, which is available in the backend the items and data, which are entered in the input, form id directly stored in this table using linking of database. We can link more than one table to input forms. We can collect the details from the different tables to display on the output.

There are mainly 12 tables in the project.They are,

1. tbllogin
2. tblcategory
3. tblward
4. tblsupervisor
5. tblanganwadi
6. tblteacher
7. tblfood
8. tblfoodcategory
9. tblcandidate
10. tblstockallocation

11. tblphysicalcondition

12. tblfoodallocation

1. Table Name : **tbllogin**

Description : This table is used to store user login information.

Table 3.1 tbllogin

Field Name	Data Type	Constraint	Description
loginid	Int	Primary Key	This field is used to uniquely identify the admin, and the values are automatically generated.
username	Varchar(50)	Not Null	This field stores the username of the admin.
password	Varchar(50)	Not Null	This field stores the password of the admin.

2. Table Name : **tblcategory**

Description : This table is used to store category information.

Table 3.2 tblcategory

Field Name	Data Type	Constraint	Description
categoryid	Int	Primary key	This field is used to uniquely identify the category, and the values are automatically generated
catimage	Varchar(200)	Not Null	This field stores the image of the category.
categoryname	Varchar(50)	Not Null	This field stores the name of the category.
description	Varchar(200)	Not Null	This field stores the description about the category.

3. Table Name : **tblward**

Description : This table is used to store the ward information.

Table 3.3 tblward

Field Name	Data Type	Constraint	Description
wardid	Int	Primary Key	This field is used to uniquely identify ward, and the values are automatically generated.
wardno	Int	Not null	This field stores the ward number.

4. Table Name : tblsupervisor

Description : This table is used to store the supervisor information.

Table 3.4 tblsupervisor

Field Name	Data Type	Constraint	Description
superid	Int	Primary Key	This field is used to uniquely identify supervisor, and the values are automatically generated.
supervisorname	Varchar(50)	Not Null	This field stores the name of the supervisor.
houasename	Varchar(100)	Not Null	This field stores the housename of the supervisor.
dob	Date	Not Null	This field stores the date of birth of supervisor.
gender	Varchar(20)	Not Null	This field stores the gender of the supervisor.
email	Varchar(50)	Not Null	This field stores the email of the supervisor.
phone	Bigint(50)	Not Null	This field stores the contact number of the supervisor.
wardid	Int	Foreign Key	This field is used to fetch the data from tblward.
username	Varchar(50)	Not Null	This field stores the username of the supervisor.
password	Varchar(50)	Not Null	This field stores the password of the supervisor.

5. Table Name : tblanganwadi

Description : This table is used to store the Anganwadi information.

Table 3.5 tblanganwadi

Field Name	Data Type	Constraint	Description
angid	Int	Primary Key	This field is used to uniquely identify anganwadi, and the values are automatically generated.
angname	Varchar(50)	Not Null	This field stores the anganwadi name.
regno	Int	Not Null	This field stores the registration number of the anganwadi.
contact	Bigint(20)	Not Null	This field stores the contact number of the anganwadi.
wardid	Int	Foreign Key	This field is used to fetch the data from tblward.

6. Table Name : tblteacher

Description : This table is used to store the Teacher information.

Table 3.6 tblteacher

Field Name	Data Type	Constraint	Description
teaid	Int	Primary Key	This field is used to uniquely identify teacher, and the values are automatically generated.
teachername	Varchar(50)	Not Null	This field stores the name of the teacher.
houasename	Varchar(100)	Not Null	This field stores the housename of the teacher.
dob	Date	Not Null	This field stores the date of birth of the teacher.
gender	Varchar(20)	Not Null	This field stores the gender of the teacher.
email	Varchar(50)	Not Null	This field stores the email of the teacher.
contact	Bigint(50)	Not Null	This field stores the contact number of the teacher.
angid	Int	Foreign Key	This field is used to fetch the data from tblanganwadi.
username	Varchar(50)	Not Null	This field stores the username of the teacher.
password	Varchar(50)	Not Null	This field stores the password of the teacher.

7. Table Name : tblfood

Description : This table is used to store the food information.

Table 3.7 tblfood

Field Name	Data Type	Constraint	Description
foodid	Int	Primary Key	This field is used to uniquely identify food, and the values are automatically generated.
fooditem	Varchar(50)	Not Null	This field stores the food item.
photo	Varchar(100)	Not Null	This field stores the food image.

8.Table Name : tblfoodcategory

Description : This table is used to store the categorywise food information.

Table 3.8 tblfoodcategory

Field Name	Data Type	Constraint	Description
foodcategoryid	Int	Primary key	This field is used to uniquely identify foodcategory, and the values are automatically generated.
categoryid	Int	Foreign Key	This field is used to fetch the data from tblcategory.
foodid	Int	Foreign Key	This field is used to fetch the data from tblfood.
foodquantity	Varchar(50)	Not Null	This field stores the quantity of food.

9.Table Name : tblcandidate

Description : This table is used to store the candidate information.

Table 3.9 tblcandidate

Field Name	Data Type	Constraint	Description
canid	Int	Primary key	This field is used to uniquely identify candidate, and the values are automatically generated.
candidatename	Varchar(50)	Not Null	This field stores the name of the candidate.
housesname	Varchar(50)	Not Null	This field stores the house name of the candidate.
dob	Date	Not Null	This field stores the date of birth of the candidate.
gender	Varchar(20)	Not Null	This field stores the gender of the candidate.
phone	Bigint(20)	Not Null	This field stores the contact number of the candidate.
email	Varchar(40)	Not Null	This field stores the email of the candidate.
aadharno	Bigint(50)	Not Null	This field stores the aadhar number of the candidate.
birthcertificate	Varchar(200)	Not Null	This field stores the birth certificate of the candidate.
height	Int	Not Null	This field stores the height of the candidate.
weight	Int	Not Null	This field stores the weight of the candidate.
categoryid	Int	Foreign Key	This field is used to fetch the data from tblcategory.

angid	Int	Foreign Key	This field is used to fetch the data from tblanganwadi.
status	Varchar(40)	Not Null	This field stores the status of the candidate.

10. Table Name : **tblstockallocation**

Description : This table is used to store the stock allocation information.

Table 3.10 tblstockallocation

Field Name	Data Type	Constraint	Description
stockid	Int	Primary Key	This field is used to uniquely identify stocks, and the values are automatically generated.
angid	Int	Foreign Key	This field is used to fetch the data from tblanganwadi
categoryid	Int	Foreign Key	This field is used to fetch the data from tblcategory
foodcategoryid	Int	Foreign Key	This field is used to fetch the data from tblfoodcategory.
quantityallocated	Vachar(50)	Not Null	This field stores the allocated quantity.
allocationdate	Date	Not Null	This field stores the allocation date.
balancequantity	Varchar(50)	Not Null	This field stores the balance food quantity.

11. Table Name : **tblphysicalcondition**

Description : This table is used to store the physical condition information.

Table 3.11 tblphysicalcondition

Field Name	Data Type	Constraint	Description
physicalconditionid	Int	Primary Key	This field is used to uniquely identify physical condition, and the values are automatically generated.
canid	Int	Foreign Key	This field is used to fetch the data from tblcandidate
height	Int	Not Null	This field stores the height of the candidate.
weight	Int	Not Null	This field stores the weight of the candidate.
checkupdate	Date	Not Null	This field stores the chekup date of the candidate.

12. Table Name : **tblfoodallocation**

Description : This table is used to store the food allocation information.

Table 3.11 tblfoodallocation

Field Name	Data Type	Constraint	Description
foodallocationid	Int	Primary Key	This field is used to uniquely identify allocated food, and the values are automatically generated.
canid	Int	Foreign Key	This field is used to fetch the data from tblcandidate
foodcategoryid	Int	Foreign Key	This field is used to fetch the data from tblfoodcategory
suppliedquantity	Varchar(50)	Not Null	This field stores the supplied quantity of food.
foodallocationdate	Date	Not Null	This field stores the date of food allocation.
status	Varchar(20)	Not Null	This field stores the status of food allocation.

3.4 DATA FLOW DIAGRAM

3.4.1 INTRODUCTION TO DATA FLOW DIAGRAMS

Data Flow Diagram is a network that describes the flow of data and processes that change, or transform, data throughout the system. This network is constructed by use a set of symbols that do not imply a physical implementation. It is a graphical tool for structured analysis of the system requirements. DFD models a system by using external entities from which data flows to a process, which transforms the data and creates, output-data-flows which go to other processes or external entities or files. Data in files may also flow to processes as inputs.

There are various symbols used in a DFD. Bubbles represent the processes. Named arrows indicate the data flow. External entities are represented by rectangles. Entities supplying data are known as sources and those that consume data are called sinks. Data are stored in a data store by a process in the system. Each component in a DFD is labelled with a descriptive name. Process names are further identified with a number.

The Data Flow Diagram shows the logical flow of a system and defines the boundaries of the system. For a candidate system, it describes the input (source), outputs (destination), database (files) and procedures (data flow), all in a format that meet the user's requirements.

The main merit of DFD is that it can provide an overview of system requirements, what data a system would process, what transformations of data are done, what files are used, and where the results flow.

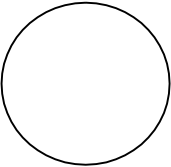
This network is constructed by use a set of symbols that do not imply a physical implementation. It is a graphical tool for structured analysis of the system requirements. DFD models a system by using external entities from which data flows to a process, which transforms the data and creates, output-data-flows which go to other processes or external entities or files. External entities are represented by rectangles. Entities supplying data are known as sources and those that consume data are called sinks. Data are stored in a data store by a process in the system. It is a graphical tool for structured analysis of the system requirements. DFD models a system by using external entities from which data flows to a process, which transforms the data and creates, output-data-flows which go to other processes or external entities or files. Data in files may also flow to processes as inputs.

Rules for constructing a Data Flow Diagram

1. Arrows should not cross each other
2. Squares, circles and files must bear names.
3. Decomposed data flow squares and circles can have same time.
4. Choose meaningful names for data flow.
5. Draw all data flows around the outside of the diagram.

Basic Data Flow Diagram Symbols

Table 3.12 Data Flow Diagram Symbols

	A data flow is a route, which enables packets of data to travel from one point to another. Data may flow from a source to a process and from data store or process. An arrow line depicts the flow, with arrow head pointing in the direction of the flow.
	Circles stands for process that converts data in to information. A process represents transformation where incoming data flows are changed into outgoing data flows.
	A data store is a repository of data that is to be stored for use by a one or more process may be as simple as buffer or queue or sophisticated as relational database. They should have clear names. If a process merely uses the content of store and does not alter it, the arrowhead goes only from the store to the process. If a process alters the details in the store, then a double-headed arrow is used.
	A source or sink is a person or part of an organization, which enters or receives information from the system, but is considered to be outside the contest of data flow model.

3.4.2 DATA FLOW DIAGRAM

Each component in a DFD is labelled with a descriptive name. Process name are further identified with number. Context level DFD is draw first. Then the process is decomposed into several elementary levels and is represented in the order of importance. A DFD describes what data flow (logical) rather than how they are processed, so it does not depend on hardware, software, and data structure or file organization.

A DFD methodology is quite effective; especially when the required design.

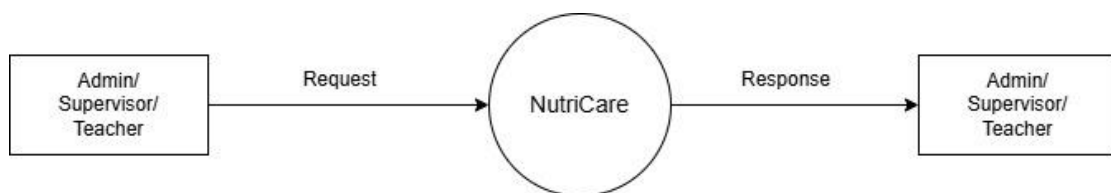


Fig. 3.1 Zeroth Level DFD for NutriCare

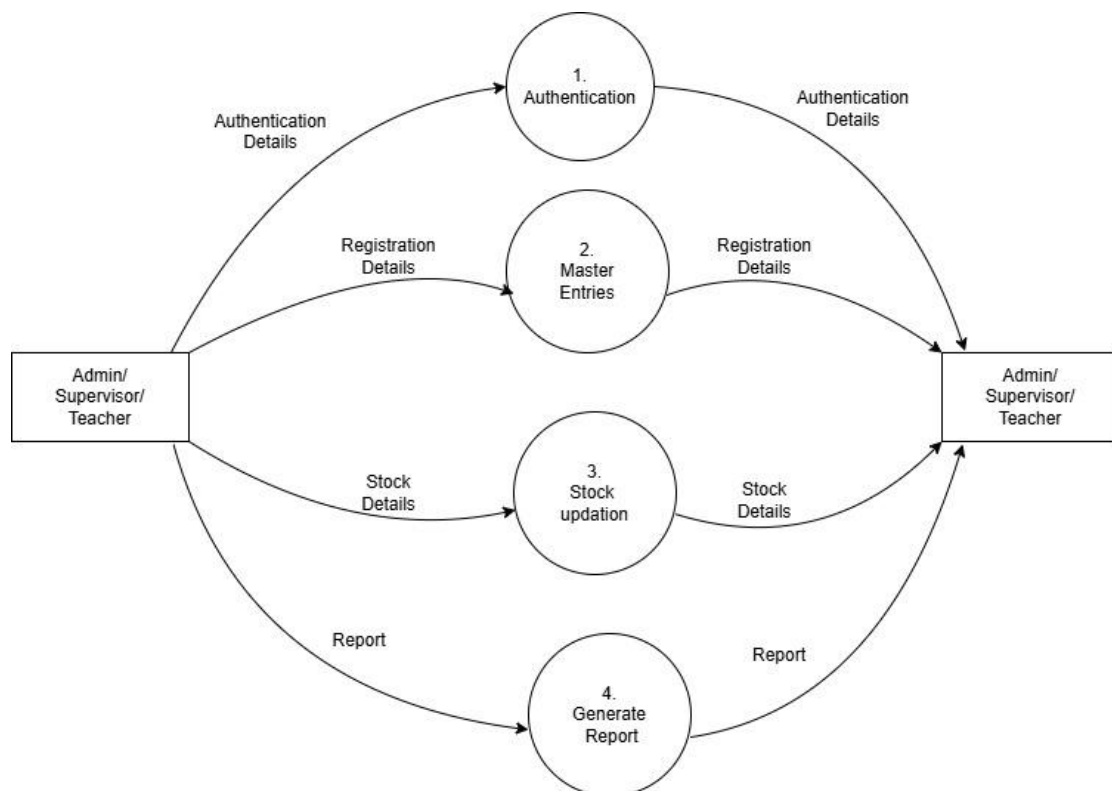


Fig. 3.2 First level DFD for NutriCare

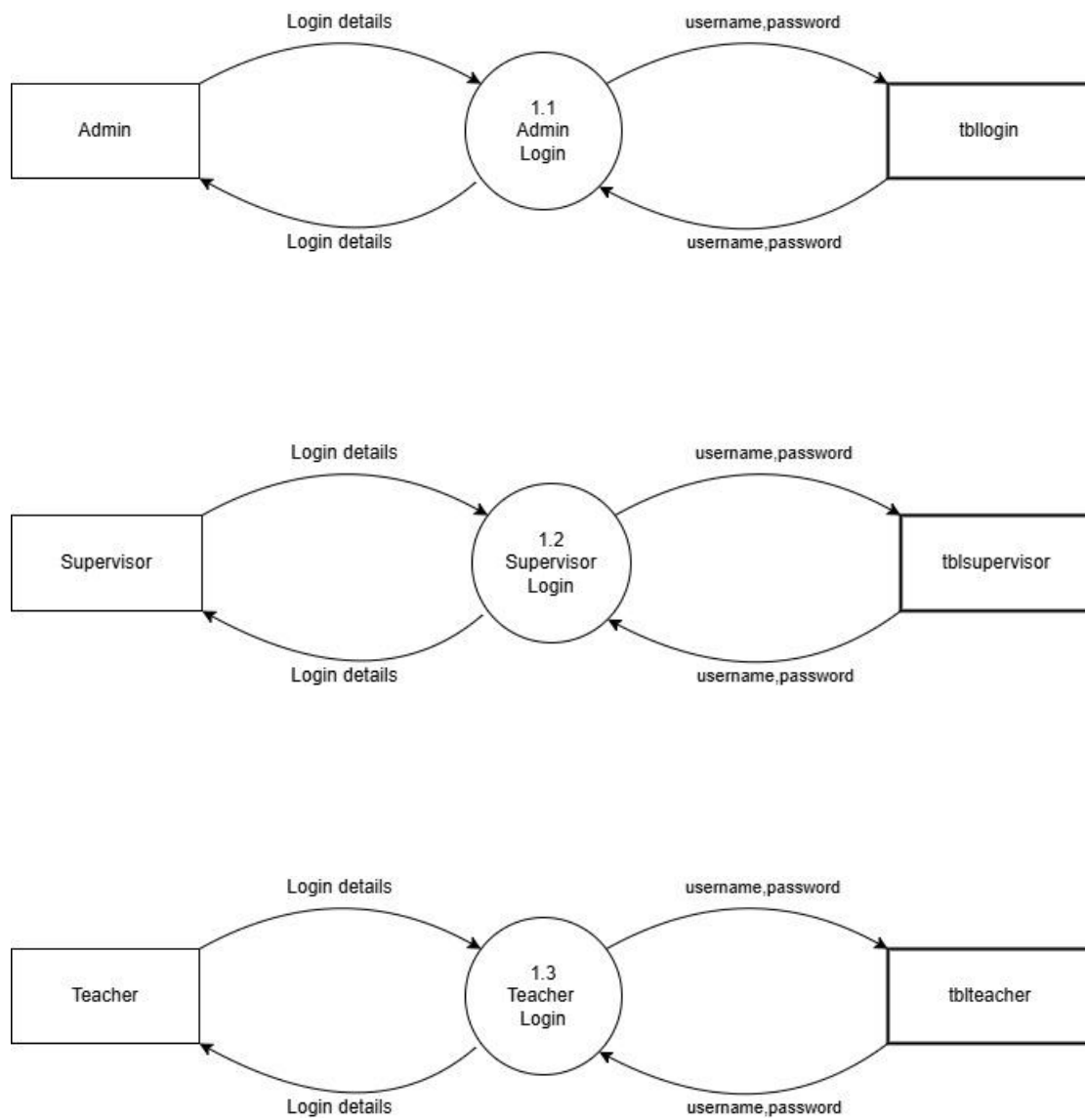
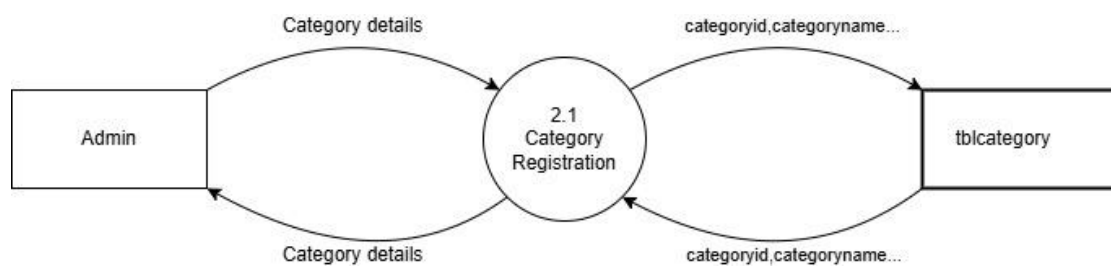
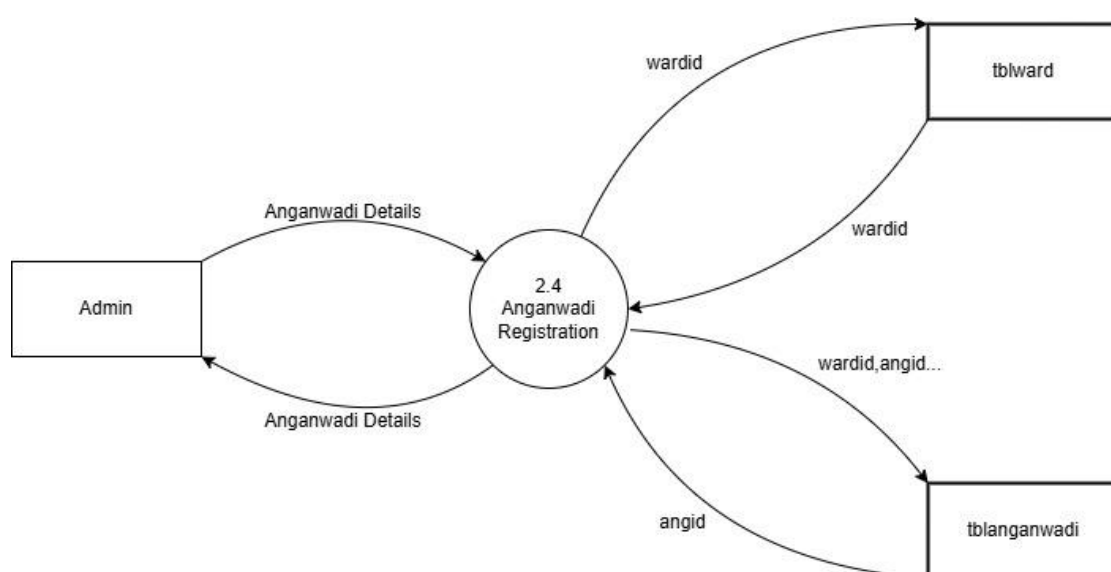
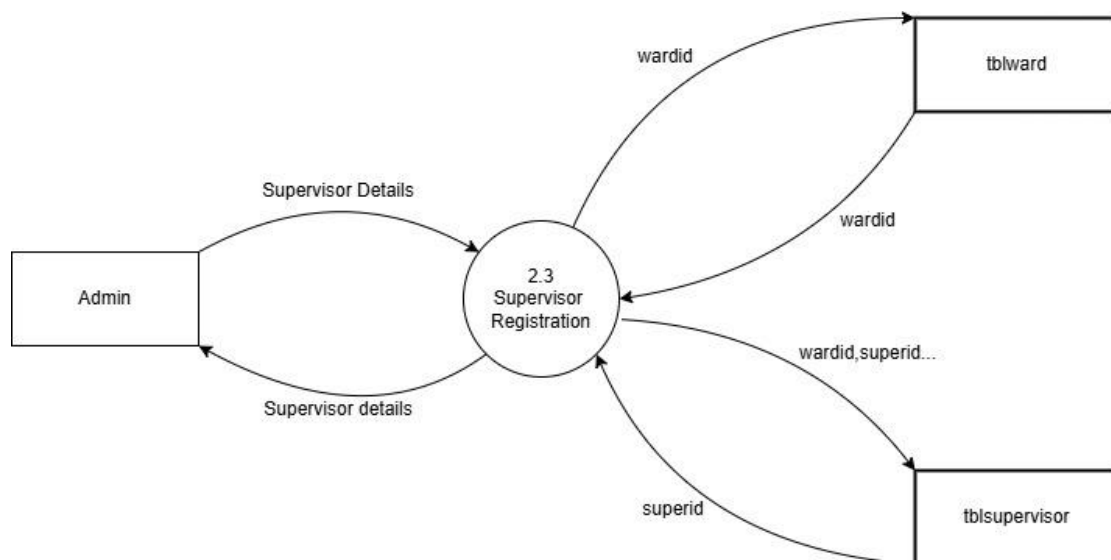
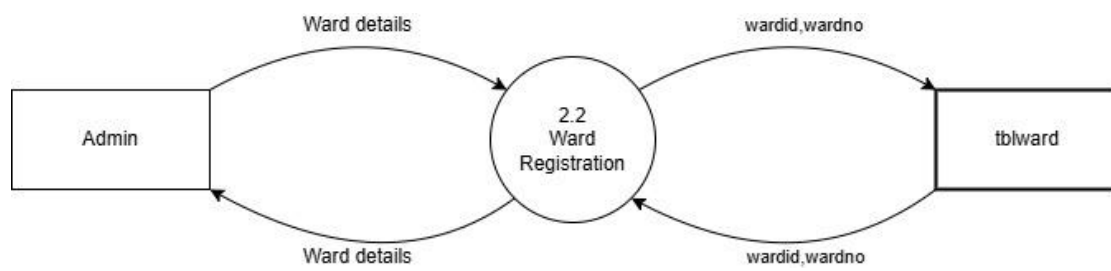
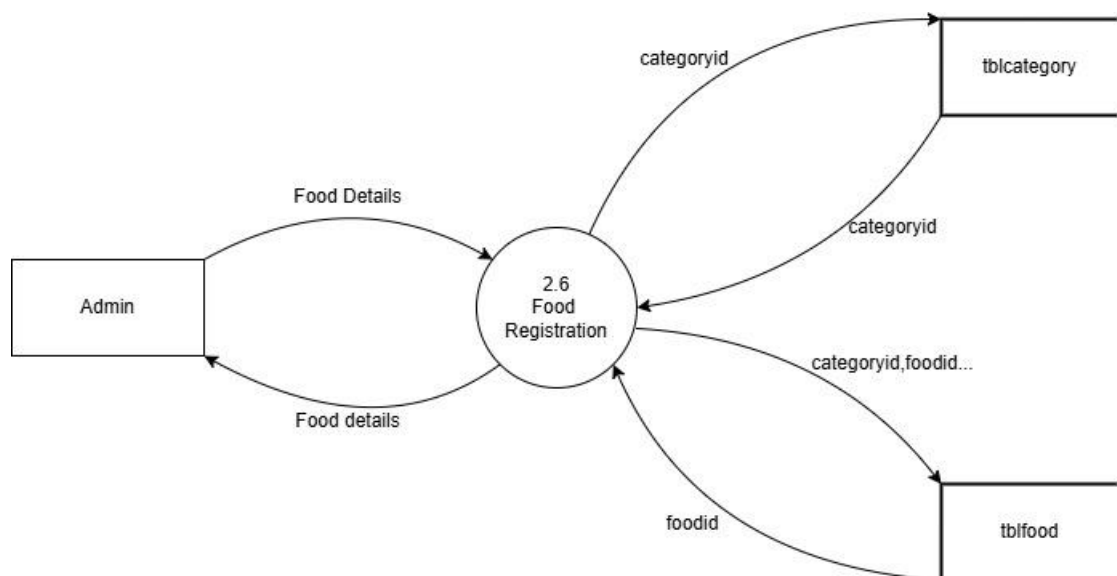
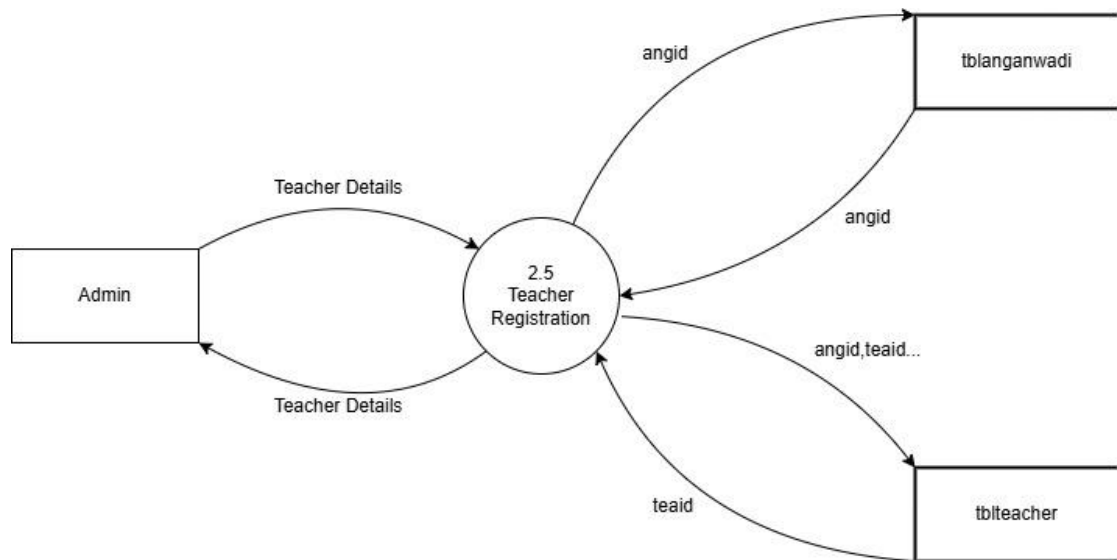


Fig. 3.3 Second Level DFD for Authentication







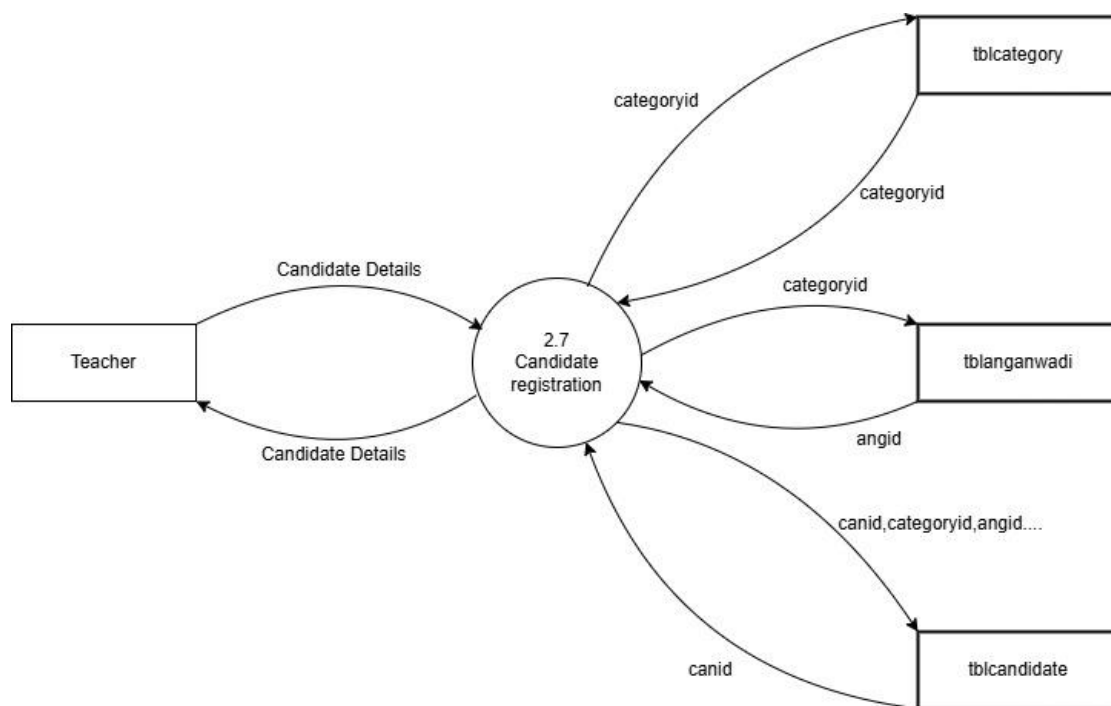


Fig. 3.4 Second Level DFD for Master Entries